Gradient Descent Method

Introduction

- **When we build a model, we find a model which**
  - Minimizes the error, or
  - Maximizes the probability

- **We have to solve one of those**

$$\arg\min_{m} E(m)$$

$$\arg\max_{m} P(m \mid D)$$

$$\arg\max_{m} P(D \mid m)$$

How to solve those??

Introduction

- **How to solve?**

$$\arg\min_{w} \; f(w)$$

  - Solution 1: Solve the following equation

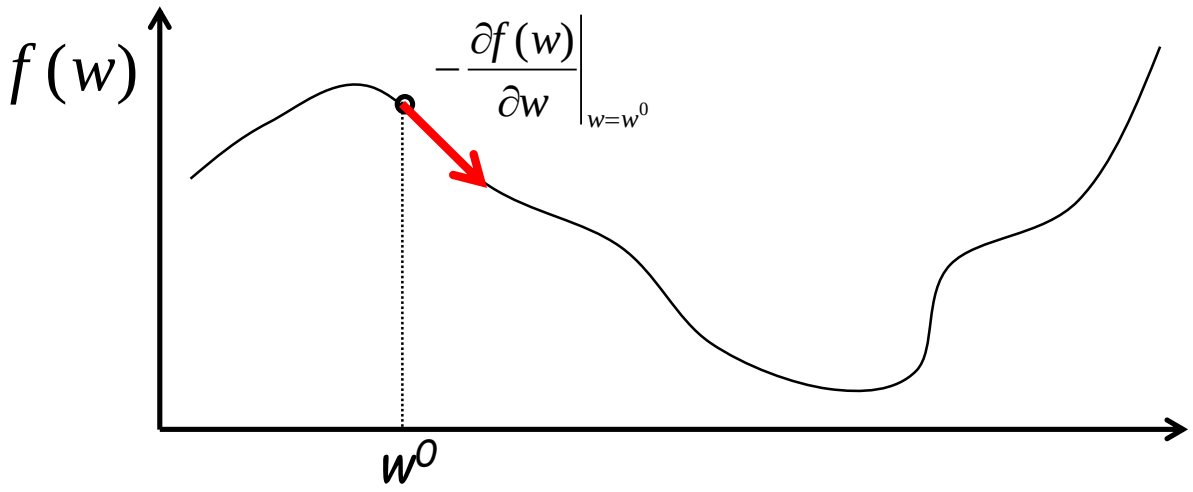
$$\frac{\partial}{\partial w} f(w) = 0$$

    - What we have done previously

  - Question: what if the equation is very difficult to solve?
    - We can adopt Gradient Descent Approach.

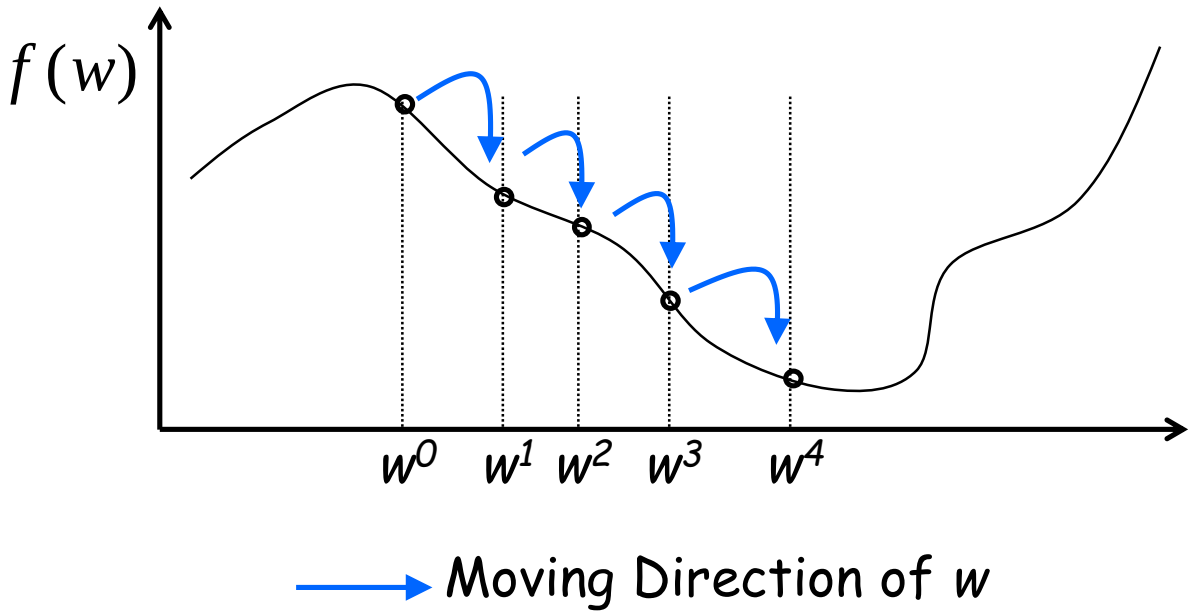
Gradient Descent Method

- **Steps**
  - Randomly choose an initial solution
  - Descend using the gradient information



Gradient Descent Method

- **Steps**
  - Randomly choose an initial solution
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Gradient Descent Method

- **Steps**

Randomly choose an initial solution, w^0

Repeat

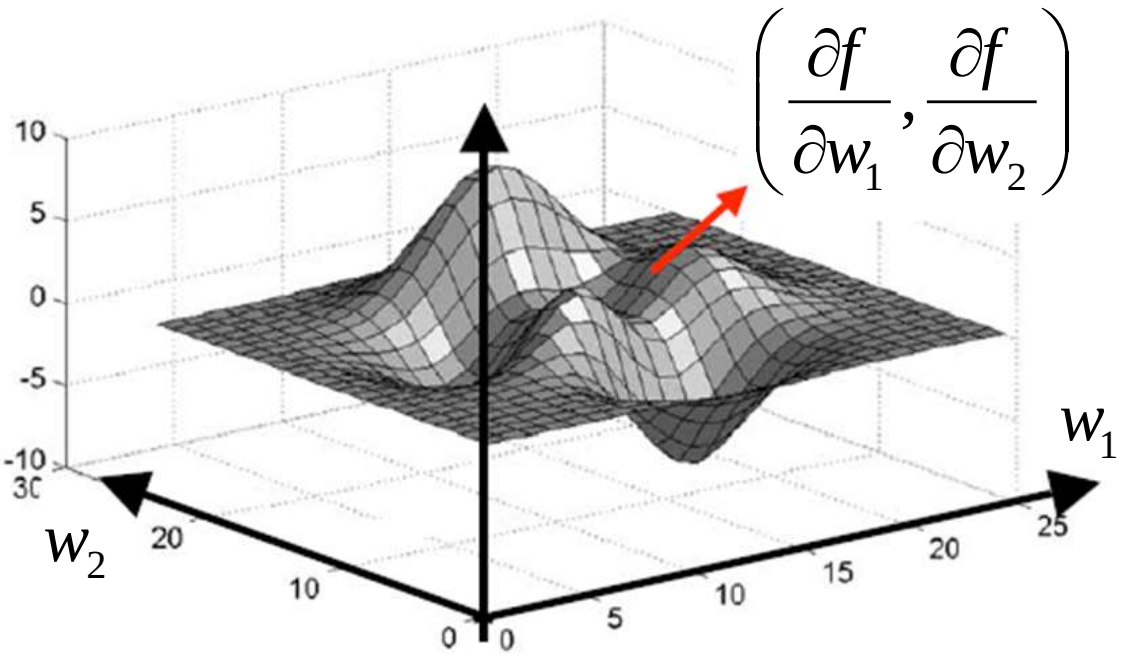
$$w^{t+1} = w^t - f'(w^t) \times \eta$$ → learning rate

Until stopping condition is satisfied

- $|w^{t+1} - w^t|$ is very small
- f(w) little moves
- fixed number of iterations

Gradient Descent Method

- **Multi-dimension**



$$\mathbf{w}^{t+1} = \mathbf{w}^t - \eta \cdot \nabla f\big|_{\mathbf{w}=\mathbf{w}^t}$$

$$\nabla f = \left(\frac{\partial f}{\partial w_1}, \frac{\partial f}{\partial w_2}\right)$$

$$w_1^{t+1} = w_1^t - \eta \cdot \frac{\partial f}{\partial w_1}\bigg|_{\mathbf{w}=\mathbf{w}^t}$$

$$w_2^{t+1} = w_2^t - \eta \cdot \frac{\partial f}{\partial w_1}\bigg|_{\mathbf{w}=\mathbf{w}^t}$$

Gradient Descent Method

- **Steps for Multi-dimension**
  - Minimize

$$f(\mathbf{w}) \text{ where } \mathbf{w} = (w_1, w_2, \cdots, w_d)$$

Randomly choose an initial solution, $\mathbf{w}^0 = (w_1^0, w_2^0, \cdots, w_d^0)$

Repeat

for j = 1 ~ d //dimension

$$w_j^{t+1} = w_j^t - \eta \cdot \left. \frac{\partial f(\mathbf{w})}{\partial w_j} \right|_{\mathbf{w}=\mathbf{w}^t}$$

Until stopping condition is satisfied

Linear Regression

- **Remind the regression problem**

Find $\mathbf{w} = (w_0, w_1, \cdots, w_d)$ which minimizes $E(\mathbf{w})$ where

$$E(\mathbf{w}) = \frac{1}{2}\sum_{i=1}^{n}\left(f(\mathbf{x}_i, \mathbf{w}) - y_i\right)^2$$

$$f(\mathbf{x}, \mathbf{w}) = w_0 x_0 + w_1 x_1 + w_2 x_2 + \cdots + w_d x_d = \sum_{i=0}^{d} w_i x_i$$

$$D = \{D_1, D_2, \cdots, D_n\}$$

$$D_i = (\mathbf{x}_i, y_i)$$

$$\mathbf{x}_i = (x_{i0}, x_{i1}, x_{i2}, \cdots, x_{id}) \text{ where } x_{i0} = 1 \text{ for } i = 1, \cdots, n$$

  - What is the target function to be minimized?

Linear Regression

- **Target function to be minimized**

$$E(\mathbf{w}) = \frac{1}{2}\sum_{i=1}^{n}\left(f(\mathbf{x}_i, \mathbf{w}) - y_i\right)^2$$

- **Let's put it into Gradient Descent Scheme!**

$$\frac{\partial}{\partial w_j}E(\mathbf{w}) = \sum_{i=1}^{n} x_{ij}\left(f(\mathbf{x}_i, \mathbf{w}) - y_i\right)$$

  - where

$$f(\mathbf{x}_i, \mathbf{w}) = w_0 x_{i0} + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_d x_{id} = \sum_{j=0}^{d} w_j x_{ij}$$

Linear Regression

- **Algorithm**

Randomly choose an initial solution, $(w_1^0, w_2^0, \cdots, w_d^0)$

Repeat

for j = 1 ~ d // dimension

$$w_j^{t+1} = w_j^t - \eta \cdot \sum_{i=1}^{n} x_{ij}\left(f\left(\mathbf{x}_i, \mathbf{w}^t\right) - y_i\right)$$

Until stopping condition is satisfied

$$\left.\frac{\partial E(\mathbf{w})}{\partial w_j}\right|_{\mathbf{w}=\mathbf{w}^t}$$

$$\frac{\partial E(\mathbf{w})}{\partial w_j} = \sum_{i=1}^{n} x_{ij}\left(f\left(\mathbf{x}_i, \mathbf{w}^t\right) - y_i\right)$$

Discussion

- **Advantage**
  - Good for optimization of a complex function of which derivative is known
  - Intuitive and easy to implement
  - Easy to parallelized

- **Disadvantage**
  - Slow
  - Give us locally optimal solution, not the optimum
    - However, for Linear Regression, it give us a solution very close to the optimal because Linear Regression is a very simple problem

Discussion

- **What happens if η is large**

Randomly choose an initial solution, $(w_1^0, w_2^0, \cdots, w_d^0)$

Repeat

for j = 1 ~ d //dimension

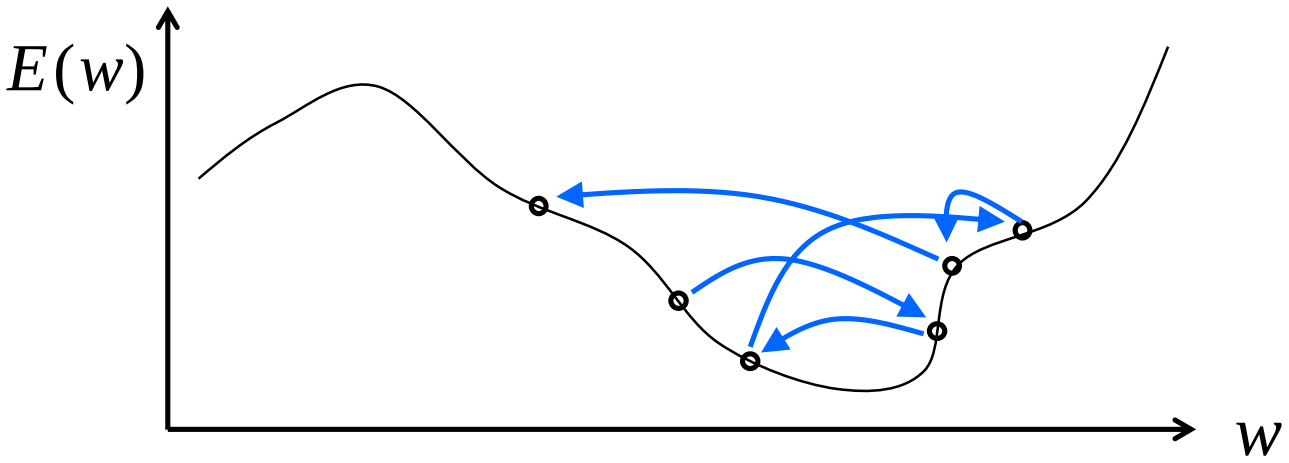
$$w_j^{t+1} = w_j^t - \eta \cdot \left. \frac{\partial E(\mathbf{w})}{\partial w_j} \right|_{\mathbf{w}=\mathbf{w}^t}$$

Until stopping condition is satisfied

  - Not converge to a local optimal
  - Jumping

Discussion

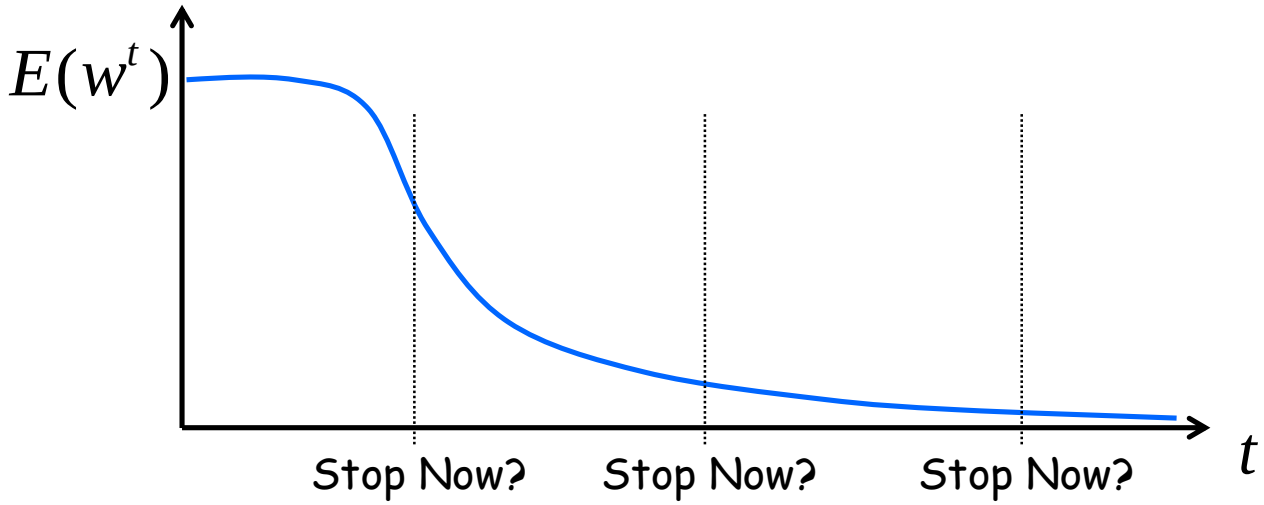
- **If η is large, your solution will wander around a local optimum**



  - Then, which η is good?

Discussion

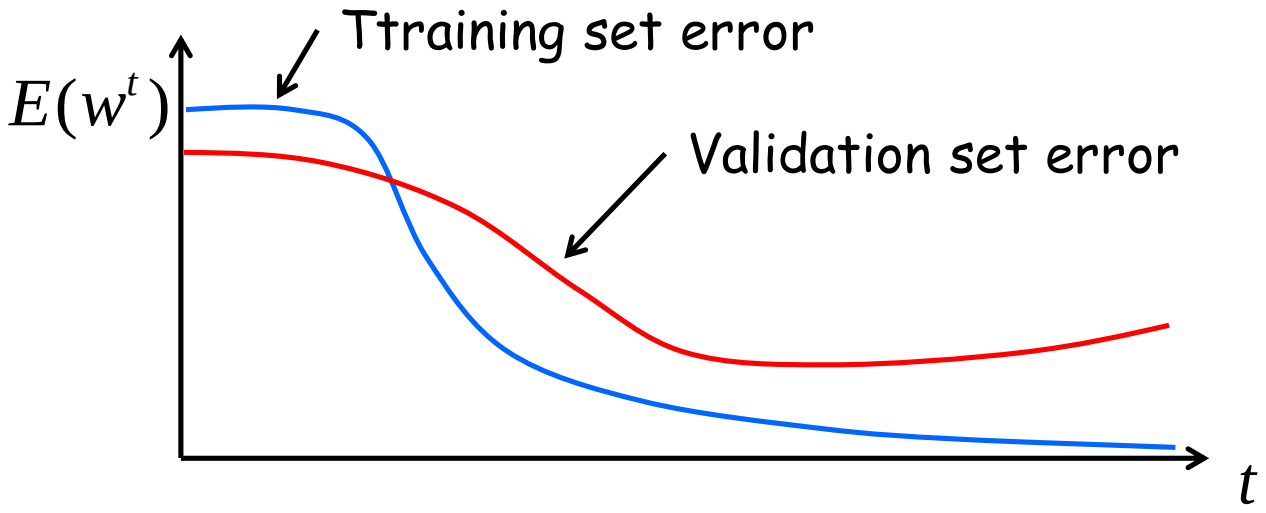
- **When to Stop**
  - Watch the error to be optimized as time goes on



  - We want to minimize *f(w)*, so…
  - Any good idea?

Discussion

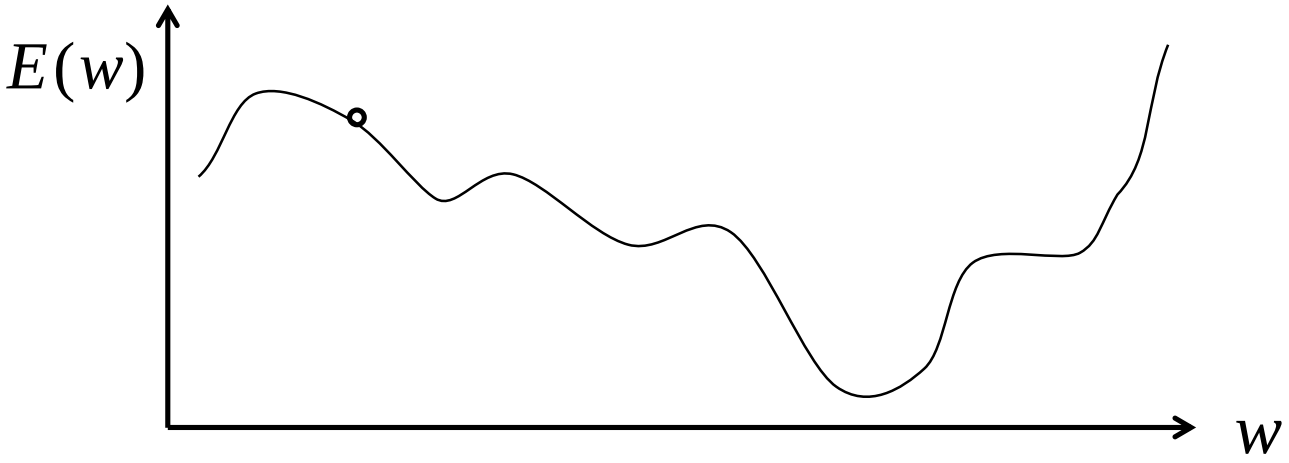
- **When to Stop**
  - Divide the training data into Training set and Validation set
  - Watch the errors for Training and Validation



  - Validation set
    - A part of training data which is used to choose parameters while learning

Discussion

- **I want to Escape from Local Optimums**
  - How?



  - Momentum

$$\Delta w_j^t = \alpha \cdot \Delta w_j^{t-1} - \eta \cdot \left. \frac{\partial E}{\partial w_j} \right|_{\mathbf{w}=\mathbf{w}^t}$$

Previously

$$\Delta w_j^t = -\eta \cdot \left. \frac{\partial E}{\partial w_j} \right|_{\mathbf{w}=\mathbf{w}^t}$$

Summary

- **Gradient Descent Method**
  - You need to determine η and # of iterations
  - Escape from local optimums: Momentum
  - Batch mode vs Online mode

- **LP by Gradient Descent Method**